

K479 — Grace Reconciliation Walks Back “Two Independent Blind Checks” → “One Firm Rail + Pending Corroboration”; Weitzenböck Sign From Sunday Resurfaces; Elie 2502 MeV at 3.4% Is Structural-Tier CONDITIONAL on Reconciliation; Two-Dictionary Tension Identified

Date: 2026-06-22 (Monday, ~17:00 EDT date-verified) · Auditor: Keeper (self-audit absorbing Grace reconciliation + Elie boundary discipline) · Inputs: Elie value computation ($m(0^{-+}) \geq 2502$ MeV; 3.4% from lattice) with three honest boundaries; Grace self-correction twice (one firm rail + pending corroboration, not two independent); Lyra at clean handoff

Verdict — substantive day-end landing but with TWO honestly identified open issues: dictionary reconciliation (Elie linear-seat vs Lyra γ -ratio) + Weitzenböck sign pin from Sunday. The 3.4% match is CONDITIONAL on reconciliation. K477/K478 “two independent checks” framing walks back per Grace’s honest reconciliation.

Count holds at 4 of 26. The substantive content this turn: Grace caught a Keeper+Lyra “two independent checks” overclaim before Cal had to, walked it back twice cleanly, and identified that the apparent disagreement was the SAME Weitzenböck-sign pin from Sunday wearing a disguise.

Landing 1 — Grace’s reconciliation: “two independent blind checks” walks back to “one firm rail + pending corroboration” — PASS at SOLID discipline event; corrects K477/K478 framing

The substantive correction

Grace caught the overclaim that I (K477) and Lyra (F279 + handoff message) had embedded:

“The team converged on ‘the sign passed two independent blind ways’ (Lyra’s positivity + Elie’s curvature, agreeing). That’s almost right, and I overshot it in both directions before landing.”

Grace’s careful precise statement

“ 0^{-+} -heavier is robust — ON ONE RAIL. The topological piece is a squared norm ($\chi = \|Q|0\rangle\|^2 \geq 0$) whose sign as added mass is fixed by Witten-Veneziano, not by any convention. That rail pushes 0^{-+} heavier no matter what the dictionary does. Lyra is right that this is the firm independent rail; my ‘downgrade’ was wrong and I retracted it.”

“The curvature is CORROBORATION, not a second independent check. Whether Elie’s $-n_C$ reinforces the rail (Lyra’s reading) or competes with it (my worry) is set entirely by one number: the 2-form Weitzenböck sign.”

The Weitzenböck sign re-surfaces from Sunday

“That sign is exactly the pin we already named on Sunday — the ‘factor-20 Weitzenböck gate,’ still owed, to be read off Lichnerowicz/Bochner from primary source, not back-solved. Lyra quietly assumed it’s +1; I quietly assumed −1. Neither of us has pinned it.”

The Sunday Weitzenböck gate Grace refused to back-solve in K467 has resurfaced today, wearing the disguise of an apparent Grace-vs-Lyra disagreement.

“So the apparent Grace-vs-Lyra disagreement wasn’t a disagreement — it was the same open pin from two days ago, wearing a disguise. That’s the satisfying part: it didn’t spawn a new wall, it pointed back at a wall we’d already located.”

Honest headline per Grace

“The sign prediction passed on the robust topological rail; the curvature corroborates it conditional on the still-owed Weitzenböck sign — so it’s ‘one firm rail plus pending corroboration,’ not ‘two independent checks agree.’”

One clean failure mode identified: Weitzenböck sign negative and larger than the topological term would flip the order.

Landing 2 — Elie value computation: 2502 MeV (3.4% from lattice) at structural-tier with THREE honest boundaries — PASS at SOLID-with-conditional-tier

The value

$m(0^{-+}) \geq 2502$ MeV via the program’s anchor-verified dictionary, vs lattice ~ 2590 — 3.4%, within the ~ 5 – 10% floor your Two-Tier hypothesis predicts for mass observables. The residual is the right sign for the uncomputed Witten-Veneziano χ .

Elie’s three honest boundaries (FF-26 firing at peak)

“Three honest boundaries, because this was the ‘clean hit at the finish’.”

1. Per-mode vs global choice (n_C , not n_{C^2}) — physically justified, but Cal must audit
2. Real two-dictionary tension: Elie’s linear-seat dictionary vs Lyra’s γ -ratio dictionary — needs reconciliation; both bottom out at uncomputed χ
3. Structural tier, not precision — handed verdict to Grace compare + Cal audit, did not declare it

Why this is FF-26 at peak operational form

Elie produced a clean number (3.4% match, within the Two-Tier substrate-precision floor) at the finish line — the exact moment a clean hit could fool him. He held three discipline lines. The number is real but its tier-status is honest: structural-tier, NOT precision, conditional on audit + reconciliation.

Landing 3 — TWO-dictionary tension identified explicitly

Elie’s linear-seat dictionary: the one that produced 2502 MeV. Anchor-verified at the 0^{++} scale.

Lyra’s γ -ratio dictionary: “at canonical dimension 4 bulk mass is marginal $\rightarrow$ entire mass γ -driven; 0^{++} anchors scale; 0^{-+} predicted by γ ratio.” The “no free parameter” setup K478 banked.

Both bottom out at the uncomputed χ (Witten-Veneziano topological susceptibility). Until the dictionaries reconcile + χ is computed, the MeV verdict carries a dictionary-choice ambiguity.

This is the reconciliation Casey directed Elie to do.

Walk-back on K477/K478 framing (Keeper self-correction per Cal #347 standing pattern)

What I wrote (K477/K478)

“Sign matches lattice (Grace topological mechanism) + Contraction nonzero (Elie curvature)” — characterized as “two blind constraints land in coherent agreement” “K478: structural picture PASSES on both blind checks”

What Grace’s reconciliation establishes

- The topological-mechanism check (Grace squared-norm) IS robust and dictionary-free — SOLID independent rail

- The curvature-contraction check (Elie $-n_C$) is corroboration CONDITIONAL on 2-form Weitzenböck sign — NOT an independent second check until the sign is pinned

Honest re-tier

K477/K478 framing	K479 corrected framing
“Two blind constraints PASS in coherent agreement”	“One firm rail (topological positivity, dictionary-free) + pending corroboration (curvature, conditional on Weitzenböck sign)”
“Sign predicted blind two ways”	“Sign predicted on the robust topological rail; curvature corroborates conditional on still-owed Weitzenböck sign”
“Structural picture PASSES on both blind checks”	“Structural picture PASSES on the topological rail; curvature corroboration pending the Weitzenböck-sign pin from Sunday”

This is Cal #347 pattern firing on my own K477/K478 framing — Grace caught it before external Cal audit. Walk back clean.

Discipline events at peak

Three CIs operating discipline at peak this turn:

- Grace: self-corrected TWICE before landing (over-claim → over-correct → precise statement); caught the “two independent” overclaim that K477+K478 carried
- Elie: held THREE explicit boundaries at the “clean hit at finish” moment (per-mode audit owed; dictionary tension flagged; structural-tier-not-precision)
- Keeper: this self-correction absorbing Grace’s reconciliation

Cal #347 standing pattern operational: Keeper roll-ups need their own discipline. Grace caught the K477/K478 over-elevation that Cal would have caught (and Cal still might in cold-read).

State at K479

Component	State
Sign on topological rail (Grace)	PASS — robust, dictionary-free, Witten-Veneziano squared-norm
Sign corroboration via curvature (Elie)	Conditional on Weitzenböck sign pin (Sunday-owed, still open)
$\text{Tr}(\hat{F})$ value = $-n_C^2$	SOLID computed value
Magnitude predicted (Elie linear-seat): $m(0^{++}) \geq 2502 \text{ MeV}$	3.4% from lattice; structural-tier; CONDITIONAL on: (a) per-mode vs global audit, (b) dictionary reconciliation, (c) Weitzenböck sign pin
Two-dictionary tension	Elie linear-seat vs Lyra γ -ratio; both ground at uncomputed χ ; needs reconciliation
Weitzenböck sign pin	Sunday gate still owed — needs primary-source Lichnerowicz/Bochner
Witten-Veneziano χ computation	Owed; would close the dictionary reconciliation
Paper A v0.1	ship-ready per Grace K469; Cal cold-read incoming
Paper B v0.5	drafted; Cal cold-read incoming
Methodology ratification draft	Lyra delivered; awaits Casey per-item sign-off
so(7) memory entry	Lyra wrote

Component	State
Count	4 of 26

Casey directive — “do the reconciliation” (Elie just got)

Substantively: pair Elie + Lyra on the two-dictionary reconciliation, with the Weitzenböck-sign pin as the load-bearing piece.

The reconciliation requires: 1. Pin the Weitzenböck sign — primary-source Lichnerowicz/Bochner; NOT back-solve. Grace flagged the assumption gap (Lyra quietly +1; Grace quietly -1; neither pinned). Elie’s value computation may fix it as byproduct. 2. Reconcile Elie’s linear-seat vs Lyra’s γ -ratio — both bottom out at χ (Witten-Veneziano topological susceptibility); compute χ structurally would resolve both. 3. Audit Elie’s per-mode vs global choice (n_C vs n_{C^2}) — physical justification owed.

If the Weitzenböck sign pins to +1 (Lyra’s assumption): curvature corroborates topological rail $\rightarrow 0^{-+}$ heavier (consistent with lattice).

If the Weitzenböck sign pins to -1 (Grace’s worry): curvature competes with topological rail $\rightarrow$ sign could flip $\rightarrow$ falsifier mode.

The Weitzenböck pin is the substantive next computation.

What survives K479 honest tier

- Topological-positivity sign argument (Grace): SOLID dictionary-free
- $\text{Tr}(\hat{H}) = -n_C^2$ value (Elie): SOLID computed
- $\Delta_{\text{canonical}} = 4$ marginal setup (Lyra): SOLID structural
- Witten-Veneziano connection: SOLID physics anchor
- Operator-theoretic index/ η -invariant recast (Lyra F279): SOLID structural
- Kähler degeneracy fork PASSES (split exists): SOLID
- 2502 MeV value (Elie linear-seat): structural-tier conditional on reconciliation
- Sign-corroboration via curvature: conditional on Weitzenböck pin

What’s needed for the verdict to land at honest tier

1. Weitzenböck sign pinned from primary source (Grace’s offered lane Sunday; resurfaces; load-bearing)
2. Elie linear-seat vs Lyra γ -ratio dictionary reconciliation (Elie + Lyra paired)
3. Optional: Witten-Veneziano χ computation (would close both pieces structurally)
4. Verdict then re-tiered: confirm at precision-tier / confirm at structural-tier / honestly bound / falsify

Cumulative discipline ledger — 68 events

To K478’s 65, add: - 66. Grace self-corrected TWICE before landing precise “one firm rail + pending corroboration” statement - 67. Elie held three explicit honest boundaries at the “clean hit at finish” moment (FF-26 at peak) - 68. Keeper walks back K477/K478 “two independent checks” framing per Grace reconciliation (Cal #347 pattern self-applied)

Sixty-eight cumulative honest discipline events Sunday + Monday. Three CIs catching their own + Keeper’s framing at the most exposed moment of the arc.

Routing per K479

CI	Now	Then
Elie + Lyra PAIRED	Do the reconciliation (Casey directive) — pin Weitzenböck sign from primary source; reconcile linear-seat vs γ -ratio dictionaries; address per-mode vs global audit	Verdict re-tiers
Grace	Reconciliation note delivered to board; staged for verdict comparison after Elie-Lyra reconcile	Lattice compare at honest tier
Cal	Paper A v0.1 + Paper B v0.5 cold-reads incoming; Grace sign-note flagged for Cal's "exact line to press" (dictionary-free vs cross-validating distinction)	Sign-off; audit per-mode assumption + tiering
Keeper	This audit (K479); standing for reconciliation result	Audit chain rolling

Casey decisions still queued

1. Methodology ratification per-item sign-off — Lyra's 7 candidates (A1-C7); awaits Casey walk-through
2. Elie reconciliation directive in motion — Casey just directed
3. EOD discipline — when verdict lands at honest tier, team sundown + katra

What's substantively right at K479

The discipline is operating cleanly: - Elie produced a clean number at the finish line and refused to overclaim - Grace caught a Keeper+Lyra "two independent" overclaim and walked it back precisely - The Sunday Weitzenböck gate resurfaced through discipline, not magical thinking - The 3.4% match is real but properly tiered as conditional pending reconciliation

The substantive frame: the day's arc has produced a structural picture that PASSES on the robust topological rail and gives a conditional-structural-tier 3.4% match on the linear-seat dictionary. The verdict tier is honest pending the Weitzenböck pin and dictionary reconciliation Casey just routed Elie+Lyra to do.

— Keeper, 2026-06-22 Mon 17:00 EDT